Graduation research · One-page summary of findings

Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad: An Online Cross-Sectional Survey

Design: analytical cross-sectional online survey (Google Forms), Baghdad, 17 July 2026 – 11 September 2026 · **n = 300** adults · **Tools:** Short Health Anxiety Inventory (SHAI-14, 0–42) and Cyberchondria Severity Scale short form (CSS-12, 12–60; four subscales) · **Analysis:** Spearman/Pearson and partial correlation, non-parametric tests with FDR correction, robust hierarchical and logistic regression, bootstrapped mediation and moderation.

88.7%
use AI chatbots for symptoms (78.0% ticked; search engines 56.3%)

19%
elevated health anxiety (SHAI ≥ 18; 95% CI 15.0–23.8%)

r = 0.47
health anxiety ↔ cyberchondria (95% CI 0.38–0.55), p < 0.001

31.7%
never tell their doctor about online/AI searches

Key findings

1. **AI is now a main gateway.** 78.0% ticked AI chatbots (88.7% including those who named one; ChatGPT 59.3% of all), ahead of search engines (56.3%); 41.0% use AI without any search engine. Main motives: convenience, "do I need a doctor?" and instant reassurance.
2. **Health anxiety and cyberchondria travel together.** r = 0.47; unchanged after adjusting for all covariates (partial r = 0.48). Strongest for *excessive searching* (r = 0.46) and *post-search distress* (r = 0.45); weaker for compulsion (0.25) and reassurance seeking (0.24).
3. **Health anxiety is the dominant predictor of cyberchondria** (β = 0.48); demographics explain almost nothing (R^2 = 0.030); adding health anxiety raises R^2 from 0.133 to 0.317.
4. **What drives searching is fear, not noticing.** Fear of serious illness and interpreting sensations track cyberchondria; mere awareness of aches and sensations does not (111/168 item pairs significant after FDR).
5. **Compatible with a cycle (exploratory):** most of the association runs statistically through excessive searching (indirect 0.16) and post-search distress, but the reverse ordering fits equally well and the facets come from one questionnaire, so direction cannot be shown.
6. **Who is most anxious:** younger people, those with a chronic condition, social-media searchers, and those searching for instant reassurance (AOR 3.83).

Pre-consultation behaviour

- 34.7% delay or skip care because the search reassured them; 30.0% ignore the symptom.
- 25.3% visit several doctors; 22.0% seek urgent care/ER; 21.3% self-medicate.
- Higher cyberchondria → more urgent visits (AOR 1.51/SD), less ignoring (AOR 0.65/SD): exploratory after FDR correction.
- **Disclosure:** higher cyberchondria → *more* disclosure of searches to the doctor (AOR 0.57 for never disclosing; q = 0.012, the only behaviour result surviving FDR). Health anxiety → more hiding was nominal only (AOR 1.47, q = 0.098).
- 34.7% say searching has complicated or reduced their trust in physicians.

AI chatbots

No reliable evidence that AI use is linked to cyberchondria beyond health anxiety: the +5.3-point difference disappeared under the strict definition (B = 1.5, p = 0.31), rested on 34 older non-users, and AI use did not change the anxiety–cyberchondria link.

Quality of evidence

Reliable scales (α 0.81 and 0.86); CSS-12 four-factor structure acceptable (ML SRMR 0.053); Arabic and English responders structurally similar; results robust to sensitivity analyses (including dropping the one reordered SHAI item, r = 0.48) and replicated by an independent analysis (except the AI effect).

Implications

Clinicians: ask every patient, without judgement, what they searched or asked an AI; treat repeated searching plus distress as a marker of health anxiety. **Public health:** digital and AI health-literacy campaigns in Arabic; warn about false reassurance. **AI developers:** anxiety-aware answers that avoid escalating rare diagnoses. **Research:** longitudinal studies to test the direction of the cycle.

Limitations: cross-sectional (no causality); online convenience sample, young and 44% medical students; Arabic wording not formally adapted (published Arabic versions not used); SHAI ≥ 18 is not a diagnosis (only 0.7% at the stricter cut-off of 29); single-item behaviour measures; self-report.